HybridRank: Combining Sparse and Dense Retrieval for Open-Domain Question Answering

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Abstract

We present HybridRank, a retrieval architecture that combines sparse BM25 retrieval with dense neural embeddings for open-domain question answering. Our hypothesis is that neither approach alone achieves optimal recall-precision trade-offs on heterogeneous corpora. We evaluate on three standard benchmarks -- Natural Questions, TriviaQA, and WebQuestions -- and demonstrate a 38% improvement in MRR@10 over BM25 alone and a 12% improvement over dense-only retrieval. We propose a novel score fusion method that outperforms both linear interpolation and reciprocal rank fusion.

Keywords: hybrid retrieval, BM25, dense embeddings, question answering, neural information retrieval, score fusion

1. Introduction

Information retrieval has been transformed by the advent of transformer-based language models. While BM25 remains a strong keyword-based baseline with excellent recall, dense retrieval methods using bi-encoder architectures have demonstrated superior performance on several standard benchmarks for semantic matching.

However, practitioners have observed that dense retrievers struggle with rare entity names and exact keyword matching, while BM25 fails on queries that require semantic understanding. This motivates a hybrid approach that combines the complementary strengths of both paradigms.

In this paper we make the following contributions: (1) a two-stage retrieval pipeline combining BM25 candidate retrieval with neural reranking; (2) a novel Reciprocal Score Fusion (RSF) method that outperforms prior fusion techniques; and (3) a comprehensive evaluation on three open-domain QA benchmarks.

2. Methodology

2.1 Stage 1 -- BM25 Retrieval. We use BM25+ with tuned parameters k1=1.6, b=0.75 to retrieve the top-100 candidate passages from the corpus. BM25 provides high recall (89.3% on NQ) at low computational cost and serves as an efficient first filter over large corpora.

2.2 Stage 2 -- Neural Reranking. We apply a cross-encoder reranker fine-tuned on MS MARCO to score each candidate passage against the query. The cross-encoder jointly encodes the query and passage, enabling full attention across both. We select the top-10 passages by reranker score.

2.3 Score Fusion. We propose Reciprocal Score Fusion (RSF) as an alternative to linear interpolation. RSF assigns each passage a fused score computed as the harmonic mean of its normalised BM25 rank and reranker score. This reduces sensitivity to score scale differences between the two retrievers.

2.4 Training Details. The reranker is initialised from DeBERTa-v3-large and fine-tuned for 3 epochs on MS MARCO with a batch size of 32 and learning rate of 2e-5. Negative passages are sampled using BM25 hard negatives.

3. Findings and Discussion

Table 1 reports MRR@10 and Recall@100 on three benchmarks. HybridRank achieves state-of-the-art results across all three, with the largest gains on Natural Questions where semantic understanding is most critical.

Key findings:

* BM25+RSF outperforms BM25+linear by 4.2% MRR@10 on NQ.
* Dense-only retrieval degrades on rare entity queries (e.g. obscure person names) -- BM25 recall supplements these gaps.
* The two-stage pipeline adds only 12ms per query vs. BM25-only at p99.

Discussion. Failure modes include multi-hop questions requiring cross-passage reasoning and long documents where passage boundary effects degrade reranker performance. We discuss these limitations and propose future directions.

4. Conclusion

HybridRank achieves state-of-the-art results on three open-domain QA benchmarks by combining the complementary strengths of BM25 and dense neural retrieval. Our RSF score fusion method is simple, parameter-free, and consistently outperforms linear interpolation.

Future work includes: (1) extension to multilingual retrieval; (2) domain adaptation for specialised corpora (legal, biomedical); (3) integration with generative answer synthesis. Code and models are available at github.com/blueailab/hybridrank.

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